

Yu-Wen Chen

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Education

University of California, Berkeley Berkeley, CA, USA
Ph.D. Candidate, Electrical Engineering and Computer Sciences (GPA: 4.0/4.0) Sep. 2022 – Present
Advisor: Dr. Murat Arcak

National Taiwan University Taipei, Taiwan
M.S., Electrical Engineering (GPA: 4.0/4.0) Sep. 2017 – June 2019
Advisor: Dr. Li-Chen Fu
Thesis: Dynamic Maneuver Control of Multi-Agent System with Input Saturation

National Taiwan University Taipei, Taiwan
B.S., Electrical Engineering (GPA: 3.7/4.0) Sep. 2012 – June 2016

Research Experience

Graduate Research Assistant (Ph.D.) Sep. 2022 – Present
Arcak's Group, Advisor: Dr. Murat Arcak, University of California, Berkeley

- Published 4 first-author papers in top control venues, including IEEE Conference on Decision and Control (CDC), American Control Conference (ACC), and IEEE Control Systems Letters (L-CSS); a fifth first-author paper is under review at IEEE Transactions on Control of Network Systems (TCNS)
- Frequently attended reading groups, workshops, and seminars across control theory, game theory, and optimization theory

Graduate Research Assistant (M.S.) Sep. 2017 – Sep. 2019
Advanced Control Laboratory, Advisor: Dr. Li-Chen Fu, National Taiwan University

- Contributed to 5 top conference and journal publications; first author on 3 of them
- Nominated for the Best Master's Thesis Award by advisor and oral committee

Undergraduate Researcher – Multi-robot Formation Control Aug. 2015 – June 2017
Advanced Control Laboratory, Advisor: Dr. Ming-Li Chiang, National Taiwan University

- Studied graph theory and advanced linear algebra; presented state-of-the-art papers weekly
- Utilized Lyapunov theorem and input-to-state stability theorem to design asymptotically stable multi-agent controllers

Undergraduate Researcher – Game Theory June 2015 – June 2016
Professor Ho-Lin Chen's Lab, Advisor: Dr. Ho-Lin Chen, National Taiwan University

- Studied auction/bidding, cost-sharing, repeated games, and dynamic games
- Devised a cost-sharing mechanism for congestion relief in consultation with Professor Chen

Teaching Experience

Graduate Teaching Assistant – Optimization Models in Engineering Fall 2026
University of California, Berkeley

- Held weekly 2-hour recitation section and 1.5-hour office hours for 200+ students
- Designed and graded homework assignments, midterm, and final problem sets

Graduate Teaching Assistant – Linear System Theory Fall 2023
University of California, Berkeley

- Held weekly 2-hour recitation section and 2.5-hour office hours for 25+ graduate students
- Designed and graded 10 homework assignments, midterm, and final problem sets

Graduate Teaching Assistant – Adaptive Control Systems Spring 2019
National Taiwan University

- Guided 20+ students to accomplish Matlab simulations and final projects

Graduate Teaching Assistant – Nonlinear System Analysis Fall 2018
National Taiwan University

- Lectured stability proofs and Advanced Linear Algebra to 15+ graduate students

Work Experience

WeRide (summer internship) June 2026 – Aug. 2026
Software Engineer Intern, North San Jose, CA, USA

- Designed algorithms to optimize pickup/dropoff (PUDO) locations for autonomous vehicles
- Developed PUDO analysis tools

MediaTek Inc. May 2021 – Aug. 2022
R&D Software Engineer, Taipei, Taiwan

- Designed and maintained middleware framework for camera applications
- Resolved customer issues related to the camera middleware to fulfill QA requirements

New Era AI Robotic Inc. Nov. 2019 – May 2021
R&D Robotics Engineer, New Taipei, Taiwan

- Designed motion planning algorithms to prevent multi-DoF robotic arms from collisions in dynamic environments
- Applied 2D and nD motion planning to autonomous mobile manipulator robots via ROS and MoveIt!
- Processed 3D sensor data and 2D image pipelines to implement robot perception on the Jetson AGX Xavier platform

Invited Talks

Invited Talk – Introduction to ROS and MoveIt!

Oct. 2021

Tunghai University, Taichung, Taiwan

- Introduced ROS and MoveIt! frameworks for robot integration and manipulation

Invited Talk – 2D and nD Planning Algorithms

Dec. 2020

National Taiwan University of Science and Technology, Taipei, Taiwan

- Introduced 2D navigation and SLAM with live ROS package demonstrations
- Presented high-dimensional planning algorithms from the Open Motion Planning Library

Project Experience

Population Games for Cyber-Physical Systems: New Theory with Tools for Transportation Management under Extreme Demand

Spring 2025 – Spring 2026

US National Science Foundation grant

- Proposed a new formulation to capture complex transportation scenarios under extreme demand
- Identified conditions guaranteeing convergence under the proposed framework

Scalable Symbolic Control: Computationally Efficient Design of Feedback Control Algorithms to Satisfy Complex Requirements

Fall 2022 – Fall 2023

US National Science Foundation grant

- Developed hierarchical control techniques in which abstract models inform feedback for complex, high-fidelity models
- Solved optimization problems via the Sum-of-Squares (SOS) technique

Amazon Racing Car Project

Fall 2022 – Fall 2023

University of California, Berkeley and Amazon Industry-Academia Cooperation

- Led 3 master's students to design and implement controllers for Amazon racing cars on a hardware platform

Multi-UAV Formation Flying and Applications

Sep. 2018 – Nov. 2020

National Chung-Shan Institute of Science and Technology (NCSIST) grant

- Proposed a novel 3D multi-agent formation controller with convergence analysis for online adaptation
- Conducted massive UAV simulations in Matlab and implemented the algorithm via MAVROS for hardware verification

Feedback Algorithms Design for Combat Model

Aug. 2017 – Aug. 2018

National Chung-Shan Institute of Science and Technology (NCSIST) grant

- Designed a game-theory-based weapon-target assignment and validated it using 5+ attrition models
- Reformulated as a control problem with stability analysis to guarantee sub-optimal performance

Publications

Yu-Wen Chen

Journal Articles

1. **Yu-Wen Chen**, Nuno C. Martins, Murat Arcak (2026). “Hierarchical Learning in Multi-Layer Population Games.” *IEEE Transactions on Control of Network Systems*. (Under review)
2. **Yu-Wen Chen**, Can Kizilkale, Murat Arcak (2026). “[Target Mirror Descent: A Unifying Framework for Solving Monotone Variational Inequalities.](#)” *IEEE Control Systems Letters*.
3. **Yu-Wen Chen**, Ming-Li Chiang, Li-Chen Fu (2023). “[Three-Dimensional Maneuver Control of Multiagent Systems With Constrained Input.](#)” *IEEE Transactions on Cybernetics*.
4. Ming-Li Chiang, **Yu-Wen Chen**, Chin-Sheng Chen, Shun-Hung Tsai (2022). “[Distributed Formation Control of Multiagent Systems With Specified Order.](#)” *IEEE Transactions on Systems, Man, and Cybernetics: Systems*.

Conference Papers

1. **Yu-Wen Chen**, Nuno C. Martins, Murat Arcak (2025). “[Hierarchical Decision-Making in Population Games.](#)” *2025 IEEE 64th Conference on Decision and Control (CDC)*.
2. **Yu-Wen Chen**, Can Kizilkale, Murat Arcak (2025). “[A Two-stage Mechanism for Prioritized Trajectory Planning in Multi-Agent Systems.](#)” *2025 American Control Conference (ACC)*.
3. **Yu-Wen Chen**, Can Kizilkale, Murat Arcak (2024). “[Solving Monotone Variational Inequalities with Best Response Dynamics.](#)” *2024 IEEE 63rd Conference on Decision and Control (CDC)*.
4. **Yu-Wen Chen**, Ming-Li Chiang, Li-Chen Fu (2020). “[Formation Control with Orientation Alignment and Constrained Input.](#)” *IFAC World Congress*.
5. **Yu-Wen Chen**, Ming-Li Chiang, Li-Chen Fu (2019). “[Ordered Formation Control and Affine Transformation of Multi-Agent Systems without Global Reference Frame.](#)” *2019 American Control Conference (ACC)*.